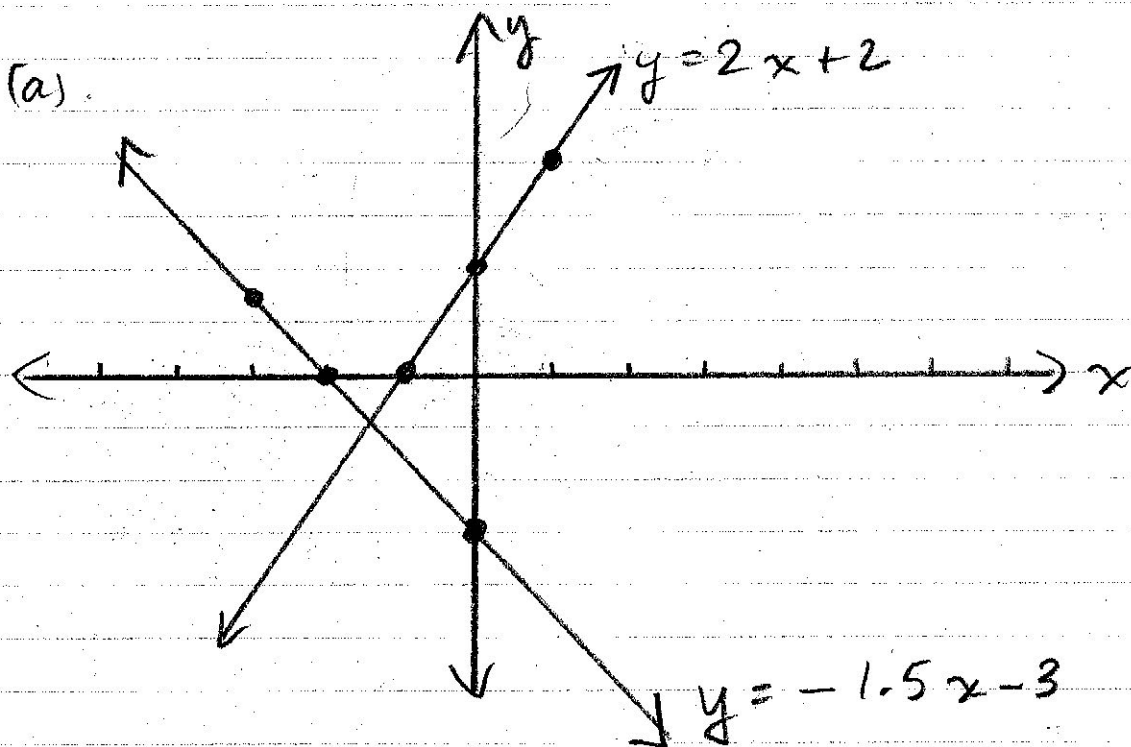
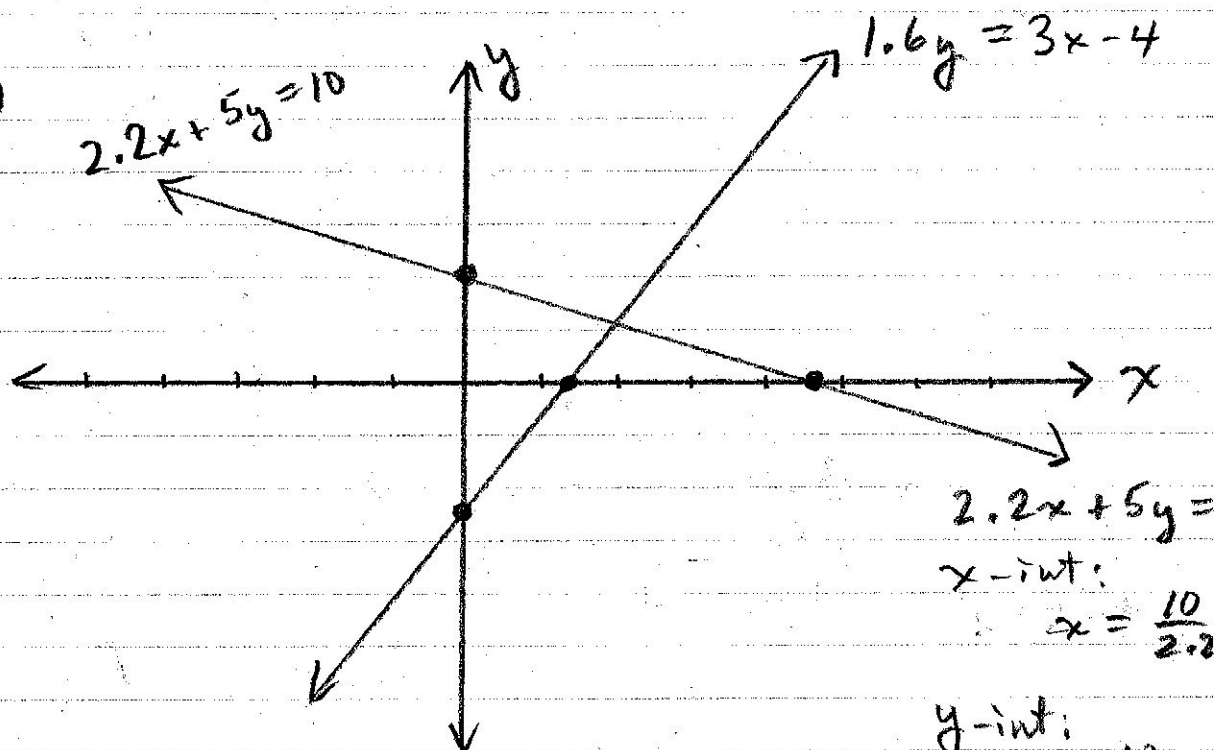


GED Math Assignment 10 Solutions

1. (a)



(b)



$$2.2x + 5y = 10$$

x-int: $x = \frac{10}{2.2}$

y-int: $y = \frac{10}{5} = 2$

$$1.6y = 3x - 4$$

x-intercept: $1.6y = -4$
 $y = -2.5$

y-int: $0 = 3x - 4$
 $x = \frac{4}{3} \approx 1.33$

$$2. (a) \quad v = \frac{\Delta d}{\Delta t} = \frac{d_2 - d_1}{t_2 - t_1}$$

$$\therefore \text{George's} \quad = \frac{250 - 25}{10 - 1}$$

$$\text{speed is} \quad = \frac{225}{9}$$

25 mph.

$$= 25 \text{ mph}$$

$$(b) \quad \Delta d = v \times t$$

$$200 = (25)(t)$$

$$t = \frac{200}{25}$$

$$= 8 \text{ h}$$

$\therefore$ it takes

George 8h to

complete the
race.

$$3. \quad y - y_1 = m(x - x_1)$$

$$y - 7 = (1)(x - 4)$$

$$\boxed{y = x + 3}$$

$$m = \frac{5 - 7}{2 - 4}$$

$$= \frac{-2}{-2}$$

$$= 1$$

4. for Safety, use rise to tread
ratio of 7:11,

$$\frac{7}{11} = \frac{6.12}{t} \rightarrow t = \frac{(11)(6.12)}{7}$$

$$= 9.61''$$

So slope = $\frac{\text{rise}}{\text{tread}}$

$$= \frac{6.12}{9.61}$$

$$= 0.64$$

5. $\frac{1}{12} = \frac{x}{12} \rightarrow x = 1 \text{ inch}$

∴ The maximum
rise is 1 inch.

6. $m = \frac{-4-0}{0-(-4)}$

$$= \frac{-4}{4}$$

$$= -1$$

So $y - 0 = (-1)(x - (-4))$

$$\boxed{y = -x - 4}$$

7. A 1. $-2 = -4(1) + b$

$$b = -2 + 4$$

$$b = 2$$

$$\text{so } y = -4x + 2$$

2. $-4 = 2(-1) + b$

$$b = -4 + 2$$

$$b = -2$$

$$\text{so } y = 2x - 2$$

3. $2 = -\frac{1}{3}(-4) + b$

$$b = 2 + \frac{4}{3}$$

$$b = \frac{10}{3}$$

$$\text{so } y = -\frac{1}{3}x + \frac{10}{3}$$

B 4. $y - 1 = 3(x - 2)$

$$y = 3x - 6 + 1$$

$$y = 3x - 5$$

5. $y - 0 = -\frac{1}{3}(x - 2)$

$$y = -\frac{1}{3}x + \frac{2}{3}$$

6. $y - (-2) = (1)(x - 1)$

$$y + 2 = x - 1$$

$$y = x - 3$$